

PARAMETER-EFFICIENT FINE-TUNING FOR 3D SCENE UNDERSTANDING

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ABSTRACT

Scene-level point cloud understanding remains challenging due to diverse geometries and spatial layouts. While pre-trained 3D point cloud foundation models (PFMs) offer strong transferability, full fine-tuning (FFT) incurs substantial computational and storage costs. Parameter-efficient fine-tuning (PEFT) provides a promising alternative, but existing PEFT methods largely focus on object-level point clouds and overlook serialization-induced partition variations in large-scale scenes. To address this issue, we propose PointPiT, a partition-invariant tuning framework for scene-level point clouds. Specifically, a Scene-aware Structural Adapter (SSA) integrates local geometric patterns with global scene context to mitigate partition-induced representation shifts. Moreover, Gradient Subspace Optimization (GSO) selects informative and partition-stable update directions, suppressing partition-dependent variations during optimization. Extensive experiments across multiple scene-level benchmarks demonstrate that PointPiT achieves competitive or even superior performance to full fine-tuning with less than 1% of backbone’s parameters, while achieving consistent state-of-the-art performance among representative PEFT methods. Our code is available at <https://github.com/gzhhhhhh/Mantis>.

Index Terms— parameter-efficient fine-tuning, point cloud, scene understanding, 3D semantic segmentation

1. INTRODUCTION

PFMs have demonstrated strong transferability across diverse 3D scene understanding tasks [1, 2, 3, 4], providing powerful pretrained representations for downstream adaptation [5, 6, 7]. Despite significant capabilities, downstream adaptation of large pretrained point cloud backbones still largely depends on FFT, incurring considerable computational and storage costs while requiring substantial task-specific supervision [8, 9, 10]. To alleviate these limitations, PEFT has emerged as an efficient alternative, achieving competitive performance with only a small fraction of trainable parameters [11, 12, 13, 14]. However, existing methods largely focus on object-level point clouds, leaving large-scale scene adaptation underexplored.

When existing PEFT methods are extended to large-scale scene-level point clouds, the potential of pretrained representations is often not fully realized. Large-scale 3D scenes comprise dense point sets with substantial geometric heterogeneity, for which serialization-based grouping is essential to tractable computation [15, 16, 17, 1]. Yet such grouping admits multiple valid partitions, exposing identical physical points to distinct local contexts. This variability can steer the limited adaptation capacity of PEFT toward partition-specific artifacts rather than task-relevant semantics.

Such partition sensitivity manifests differently across adaptation paradigms. As illustrated in Fig. 1, FFT can largely reconcile partition-induced discrepancies through unrestricted backbone updates, whereas linear probing simply inherits the partition sensitivity

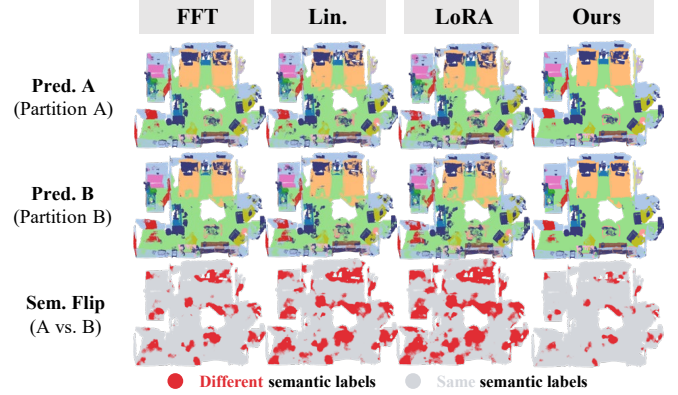


Fig. 1: Illustration of partition sensitivity on the ScanNet Val [19].

of the frozen representation. More critically, LoRA [18] adapts the backbone within a restricted update space yet lacks an explicit mechanism to distinguish task-relevant variation from partition-specific perturbations, thereby rendering its limited adaptation budget vulnerable to the latter. Such entanglement between task adaptation and partition-specific perturbations limits the effective exploitation of pretrained scene representations by existing PEFT methods.

In this paper, we propose PointPiT, a partition-invariant tuning framework tailored for large-scale scene-level point clouds. PointPiT addresses partition sensitivity from both forward representation adaptation and backward optimization. Specifically, the *Scene-aware Structural Adapter (SSA)* integrates local geometric patterns with global scene context to mitigate partition-induced representation shifts. *Gradient Subspace Optimization (GSO)* further characterizes parameter sensitivity and partition-dependent gradient variation to identify a stable update subspace, suppressing partition-sensitive components during back-propagation.

Extensive experiments demonstrate the effectiveness and efficiency of PointPiT. When integrated into Point Transformer V3 (PTv3) [1], PointPiT reduces the number of trainable parameters by more than 99% compared with full fine-tuning of the backbone, while maintaining comparable or even better performance across benchmarks. Our main contributions are summarized as follows:

- We reveal partition sensitivity as a critical challenge for PEFT in large-scale 3D scenes, exposing the mismatch between scene serialization and parameter-efficient adaptation.
- We propose PointPiT, where SSA reduces partition-induced representation shifts and GSO restricts optimization to partition-stable gradient directions.
- Extensive experiments on multiple scene-level benchmarks demonstrate that PointPiT achieves SOTA performance among PEFT methods, while attaining performance comparable to FFT with substantially fewer trainable parameters.

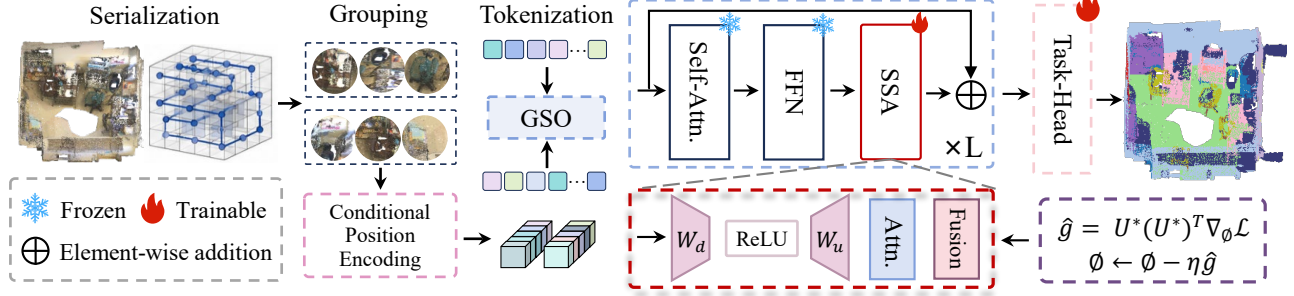


Fig. 2: Overview of PointPiT. Serialized scene points are grouped, position encoded, and embedded into tokens before entering the frozen backbone. PointPiT consists of a Scene-aware Structural Adapter (SSA) for forward representation adaptation and Gradient Subspace Optimization (GSO) for backward optimization. SSA is inserted into each backbone block to enhance scene-level structural representations, while GSO constrains gradient updates to partition-stable directions. During training, only SSA and the task head are updated.

2. METHODOLOGY

2.1. Overview

Given a large-scale point cloud scene $\mathbf{X} \in \mathbb{R}^{N \times d}$, where N denotes the number of points and d represents the input feature dimension, PointPiT aims to efficiently adapt pretrained point cloud foundation models to downstream scene understanding tasks. Building on the pioneer works [1], we define the point tokenization process as $\mathcal{T} = \mathcal{E} \circ \mathcal{G} \circ \mathcal{S}$, where $\mathcal{S}(\cdot)$ transforms unordered points into ordered sequences, $\mathcal{G}(\cdot)$ partitions serialized points into local groups, and $\mathcal{E}(\cdot)$ projects grouped points into feature tokens. The initial token representation is therefore obtained as $\mathbf{H}^0 = \mathcal{T}(\mathbf{X})$. The pretrained backbone consists of L hierarchical blocks. Specifically, the forward process of each block is defined as:

$$\mathbf{H}^{l+1} = \mathcal{B}^l(\mathbf{H}^l), \quad l = 0, 1, \dots, L-1, \quad (1)$$

where $\mathcal{B}^l$ denotes the l -th backbone block. To enable efficient adaptation for large-scale scenes, PointPiT introduces two complementary components. The proposed *Scene-aware Structural Adapter* (SSA) enhances forward representation adaptation by incorporating both local geometric patterns and global scene context. Meanwhile, *Gradient Subspace Optimization* (GSO) optimizes the backward update trajectory by preserving task-relevant adaptation directions.

2.2. Scene-aware Structural Adapter

Large-scale scenes exhibit diverse geometric structures, where serialized local groups may suffer from limited contextual perception due to restricted receptive fields. To improve scene-level adaptation, SSA jointly models local geometric patterns and global scene context for robust adaptation.

Given the intermediate representation $\mathbf{H}^l \in \mathbb{R}^{N_l \times C_l}$, SSA first extracts local adaptation features through a lightweight bottleneck projection:

$$\mathbf{H}_{loc}^l = W_u^l \sigma(W_d^l \mathbf{H}^l), \quad (2)$$

where $W_d^l \in \mathbb{R}^{C_l \times r}$ and $W_u^l \in \mathbb{R}^{r \times C_l}$ denote the projection matrices, r represents the hidden dimension, and $\sigma(\cdot)$ denotes the nonlinear activation function.

To incorporate global scene-level structural information, SSA aggregates the input representation into a scene descriptor $\mathbf{z}^l = \text{AvgPool}(\mathbf{H}^l)$, and performs global-local interaction:

$$\mathbf{H}_{ctx}^l = \text{Softmax} \left(\frac{\mathbf{H}^l (W_q^l \mathbf{z}^l)^T}{\sqrt{C_l}} \right) W_v^l \mathbf{H}^l, \quad (3)$$

where W_q^l and W_v^l denote the query and value projection matrices, respectively. The resulting contextual feature captures long-range dependencies beyond individual local groups.

Finally, the local and contextual representations are adaptively fused through a learnable gating mechanism:

$$\tilde{\mathbf{H}}^l = \mathbf{H}^l + \mathbf{g}^l \odot \mathbf{H}_{loc}^l + (1 - \mathbf{g}^l) \odot \mathbf{H}_{ctx}^l, \quad (4)$$

where $\mathbf{g}^l$ denotes the fusion coefficient generated by a lightweight gating network and $\odot$ represents element-wise multiplication.

Consequently, SSA establishes a structural adaptation pathway that reduces partition-induced representation shifts, while providing stable feature spaces for GSO to optimize gradient updates.

2.3. Gradient Subspace Optimization

Although SSA stabilizes forward representations, backward optimization remains susceptible to partition-induced gradient variations. Given the trainable parameters $\phi \in \mathbb{R}^{D_\phi}$, the optimization gradient can be conceptually decomposed as $\nabla_{\phi} \mathcal{L} = \mathbf{g}_t + \mathbf{g}_p$, where $\mathbf{g}_t$ and $\mathbf{g}_p$ denote the partition-consistent and partition-induced components, respectively.

To characterize informative update directions during optimization, we construct the empirical Fisher information matrix over a mini-batch of B training scenes:

$$\mathbf{F} = \frac{1}{B} \sum_{b=1}^B \nabla_{\phi} \mathcal{L}(\mathbf{X}_b) [\nabla_{\phi} \mathcal{L}(\mathbf{X}_b)]^T, \quad (5)$$

where $\mathbf{X}_b$ denotes the b -th training scene and $\mathbf{F} \in \mathbb{R}^{D_\phi \times D_\phi}$ characterizes the sensitivity of the objective to different parameter-update directions.

Given K valid partitions $\{\mathcal{P}_k\}_{k=1}^K$ of the same physical scene, the corresponding gradients are represented as $\mathbf{g}_k = \nabla_{\phi} \mathcal{L}(\mathcal{P}_k)$. We characterize partition-dependent gradient variation through

$$\mathbf{G}_p = \frac{1}{K} \sum_{k=1}^K (\mathbf{g}_k - \bar{\mathbf{g}}) (\mathbf{g}_k - \bar{\mathbf{g}})^T, \quad (6)$$

where $\bar{\mathbf{g}} = \frac{1}{K} \sum_{k=1}^K \mathbf{g}_k$ denotes the mean gradient across partitions, and $\mathbf{G}_p \in \mathbb{R}^{D_\phi \times D_\phi}$ captures partition-induced gradient covariance. GSO seeks a q -dimensional subspace $\mathbf{U} \in \mathbb{R}^{D_\phi \times q}$, with $q \ll D_\phi$, that preserves informative optimization directions while suppressing partition-sensitive variations. The subspace is obtained by solving

$$\mathbf{U}^* = \underset{\mathbf{U}^T \mathbf{U} = \mathbf{I}_q}{\text{argmax}} \text{Tr} [\mathbf{U}^T (\mathbf{F} - \lambda \mathbf{G}_p) \mathbf{U}], \quad (7)$$

Table 1: Semantic segmentation results on ScanNet Validation [19]. We report mean accuracy (mAcc), overall accuracy (allAcc), mean IoU (mIoU), per-class IoU (%), and trainable parameters. The dec. and w/o dec. indicate Sonata with and without the decoder, respectively. Sonata (lin.) serves as the baseline for PEFT comparison. Among all PEFT methods, the best and second-best results are highlighted in black and underlined, respectively. The same notation applies to the following tables.

Method	ScanNet Val [19]			wall	floor	cabinet	bed	chair	sofa	table	door	window	books	picture	counter	desk	curtain	refrig.	shower	toilet	sink	bathtub	other
	Param.	mAcc	mIoU																				
<i>Full Fine-tuning</i>																							
PTv3-PPT (sup.) [20]	124.8	85.9	78.6	87.6	96.3	75.4	85.3	88.9	83.9	77.3	69.0	70.0	75.3	66.0	72.6	69.9	79.0	77.7	79.4	93.7	75.7	85.8	63.2
Sonata (dec.) [2]	124.8	86.1	79.4	88.3	96.3	76.3	86.3	89.0	84.8	78.4	70.2	72.2	76.2	67.4	74.1	70.4	80.1	78.8	79.8	93.6	76.6	85.4	63.8
Sonata (w/o dec.) [2]	108.5 (100%)	86.1	78.9	87.6	96.2	76.1	85.3	88.8	84.6	77.8	69.8	70.8	75.5	66.6	73.7	69.9	79.4	77.6	80.0	93.0	76.1	85.8	63.4
<i>General PEFT Methods</i>																							
Sonata (lin.)	0.02 (0.02%)	83.1	72.2	82.8	93.5	70.4	82.2	85.8	78.6	71.5	58.4	63.0	69.3	55.0	65.4	59.8	71.5	71.0	72.7	91.3	68.3	80.9	52.6
+ Adapter [21]	1.90 (1.76%)	85.1 (+2.0)	76.8 (+4.6)	86.0	97.0	74.5	84.7	87.4	82.4	77.3	68.5	68.3	72.0	62.0	69.9	67.6	76.8	75.6	78.2	91.8	72.6	84.1	59.3
+ Prefix Tuning [22]	0.08 (0.08%)	84.2 (+1.1)	73.5 (+1.3)	84.7	96.3	70.0	81.9	85.9	80.1	72.4	61.8	64.0	69.8	57.5	66.5	63.7	72.8	72.6	74.7	92.7	69.3	80.7	52.6
+ LoRA [18]	0.89 (0.82%)	85.2 (+2.1)	76.3 (+4.1)	85.5	96.8	72.6	84.5	85.9	81.9	74.3	65.5	67.8	71.1	61.9	71.2	67.1	77.8	75.9	76.7	92.7	72.8	84.3	59.7
<i>PEFT Methods for Point Cloud</i>																							
+ IDPT [8]	2.61 (2.42%)	82.9 (-0.2)	72.6 (+0.4)	84.4	95.2	69.1	81.7	85.2	79.4	71.1	60.5	62.0	68.0	57.9	65.7	61.7	72.8	70.8	73.1	91.0	69.4	81.0	52.0
+ DAPT [23]	1.14 (1.06%)	86.5 (+3.4)	77.7 (+5.5)	86.7	95.9	75.2	85.2	87.8	83.6	76.8	68.2	69.2	73.8	64.1	71.1	68.0	78.3	76.9	79.6	93.9	74.2	84.9	60.6
+ PointGST [9]	1.05 (0.97%)	85.8 (+2.7)	77.7 (+5.5)	85.7	96.4	73.9	84.9	89.3	82.5	76.9	68.3	69.0	73.8	65.4	71.5	67.9	78.2	76.9	79.5	93.8	74.4	84.3	61.4
+ PointTPA [7]	1.18 (1.09%)	86.3 (+3.2)	78.4 (+6.2)	88.0	96.6	75.3	85.9	88.5	83.5	78.1	68.2	70.3	75.1	65.1	72.6	69.0	78.9	76.3	79.6	93.7	75.2	84.5	63.6
+ PointPiT (ours)	1.08 (0.99%)	86.7 (+3.6)	78.9 (+6.7)	87.3	96.5	75.9	86.3	89.1	84.0	77.9	69.6	71.2	75.0	66.3	73.2	70.1	79.2	78.0	79.8	93.8	75.9	85.6	63.3

Table 2: Semantic segmentation results on S3DIS Area 5 [24], S3DIS 6-fold [24], ScanNet200 Validation [25], and ScanNet++ Validation [26] sets. We report mean accuracy (mAcc), overall accuracy (allAcc), and mean IoU (mIoU). Trainable parameters are consistent with Table 1.

Method	S3DIS Area 5 [24]			S3DIS 6-fold [24]			ScanNet200 Val [25]			ScanNet++ Val [26]		
	mAcc	allAcc	mIoU	mAcc	allAcc	mIoU	mAcc	allAcc	mIoU	mAcc	allAcc	mIoU
<i>Full Fine-tuning</i>												
PTv3-PPT (sup.) [20]	80.1	92.0	74.3	88.6	92.7	80.5	45.9	83.8	35.9	55.7	86.4	43.3
Sonata (dec.) [2]	81.6	93.0	76.0	89.9	93.3	82.3	46.5	84.4	36.8	55.8	86.6	43.7
Sonata (w/o dec.) [2]	80.3	93.1	74.5	87.3	92.3	79.5	47.8	83.7	37.3	50.7	86.3	41.8
<i>General PEFT Methods</i>												
Sonata (lin.)	80.9	90.8	73.0	87.4	90.8	76.5	41.6	81.2	29.3	49.4	84.9	36.5
+ Adapter [21]	81.7(+0.8)	91.6(+0.8)	73.8(+0.8)	87.5(+0.1)	91.4(+0.6)	76.4(-0.1)	45.7(+4.1)	82.5(+1.3)	33.6(+4.3)	52.9(+3.5)	86.1(+1.2)	39.9(+3.4)
+ Prefix Tuning [22]	82.5(+1.6)	91.0(+0.2)	73.4(+0.4)	86.5(-0.9)	90.5(-0.3)	73.7(-2.8)	44.4(+2.8)	81.6(+0.4)	31.4(+2.1)	49.6(+0.2)	84.2(-0.7)	36.8(+0.3)
+ LoRA [18]	81.1(+0.2)	92.5(+1.7)	74.0(+1.0)	87.8(+0.4)	91.2(+0.4)	77.4(+0.9)	45.5(+3.9)	82.7(+1.5)	33.6(+4.3)	51.4(+2.0)	85.3(+0.4)	38.7(+2.2)
<i>PEFT Methods for Point Cloud</i>												
+ IDPT [8]	81.2(+0.3)	90.5(-0.3)	72.0(-1.0)	86.8(-0.6)	90.6(-0.2)	75.2(-1.3)	41.9(+0.3)	81.0(-0.2)	29.0(-0.3)	50.0(+0.6)	84.1(-0.8)	36.6(+0.1)
+ DAPT [23]	81.2(+0.3)	92.7(+1.9)	74.6(+1.6)	88.1(+0.7)	91.8(+1.0)	78.2(+1.7)	45.8(+4.2)	82.9(+1.7)	34.2(+4.9)	50.9(+1.5)	85.2(+0.3)	39.6(+3.1)
+ PointGST [9]	81.4(+0.5)	92.9(+2.1)	75.0(+2.0)	88.6(+1.2)	92.2(+1.4)	78.9(+2.4)	46.5(+4.9)	83.2(+2.0)	35.0(+5.7)	52.0(+2.6)	85.9(+1.0)	40.0(+3.5)
+ PointTPA [7]	81.7(+0.8)	92.9(+2.1)	74.9(+1.9)	88.9(+1.5)	92.5(+1.7)	79.4(+2.9)	47.2(+5.6)	83.8(+2.6)	35.8(+6.5)	52.9(+3.5)	86.1(+1.2)	40.9(+4.4)
+ PointPiT (ours)	82.3(+1.4)	93.3(+2.5)	75.0(+2.0)	90.2(+2.8)	93.6(+2.8)	80.6(+4.1)	49.0(+7.4)	84.8(+3.6)	38.1(+8.8)	56.4(+7.0)	87.0(+2.1)	44.2(+7.7)

where $\text{Tr}(\cdot)$ denotes the matrix trace, λ controls the trade-off between parameter sensitivity and partition stability, and $\mathbf{I}_q$ is the q -dimensional identity matrix. The optimal $\mathbf{U}^*$ is formed by the leading eigenvectors of $\mathbf{F} - \lambda \mathbf{G}_p$.

During back-propagation, GSO removes gradient components outside the learned subspace:

$$\hat{\mathbf{g}} = \nabla_{\phi} \mathcal{L} - \left[\mathbf{I} - \mathbf{U}^*(\mathbf{U}^*)^T \right] \nabla_{\phi} \mathcal{L} = \mathbf{U}^*(\mathbf{U}^*)^T \nabla_{\phi} \mathcal{L}. \quad (8)$$

This restricts each update to the selected subspace without changing the set of trainable parameters. The parameters are then updated by $\phi \leftarrow \phi - \eta \hat{\mathbf{g}}$, where η is the learning rate. Consequently, GSO suppresses partition-dependent optimization variations and complements the representation stabilization of SSA.

3. EXPERIMENTS

3.1. Experimental Setup

Datasets. To evaluate scene-level semantic segmentation, we benchmark PointPiT on three large-scale benchmarks, including ScanNet [19], ScanNet++ [26], and S3DIS [24]. These datasets contain diverse indoor scenes with different scales and semantic categories, providing comprehensive evaluations for scene-level adaptation.

Implementation Details. We adopt PTv3 [1] initialized with pre-trained Sonata [2] weights as the backbone. Following the PEFT setting [8], the backbone parameters are frozen during training, and only the proposed SSA modules and task heads are optimized. we set the prefix [22] token length to 4, the LoRA [18] rank to 32, and the adapter’s [21] intermediate dimension to 64. The hyperparameters are shared across all datasets throughout the validation process. All experiments are conducted on 8 NVIDIA A800 GPUs.

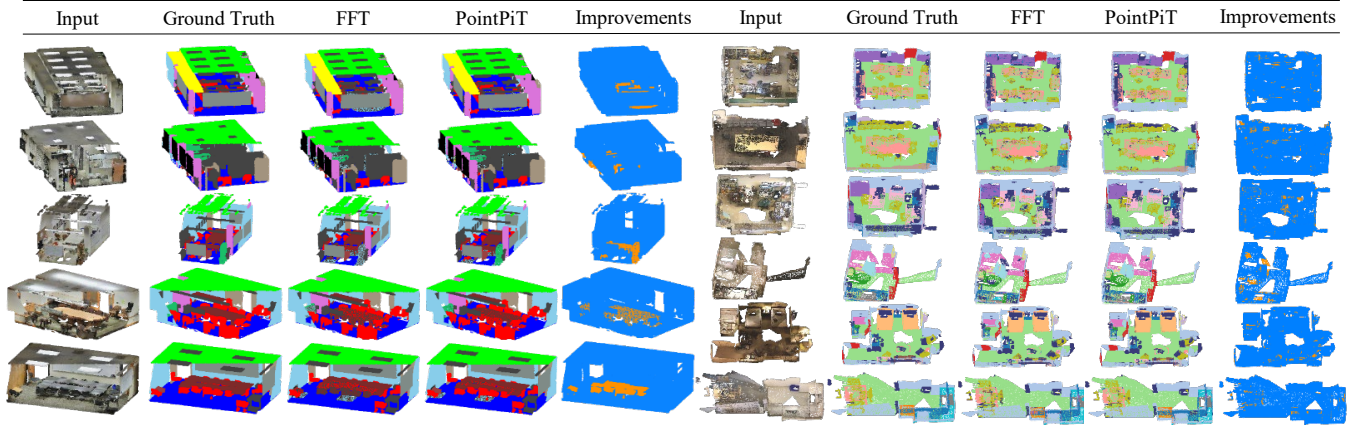


Fig. 3: Qualitative comparison of scene-level semantic segmentation. The left and right parts present results on S3DIS Area 5 [24] and ScanNet [19], respectively. Compared with FFT, PointPiT generates more accurate and consistent predictions in complex scene regions.

Table 3: Ablation study of different components of PointPiT.

SSA	GSO	ScanNet Val [19]	
		mAcc	mIoU
		83.1	72.2
✓		85.8	77.5
	✓	84.6	75.9
✓	✓	86.7	78.9

Table 4: Sensitivity to different serialization strategies. We report mIoU (%) on ScanNet Val [19].

Method	Z-order	Hilbert	Trans-Z	Trans-H
Sonata (w/o dec.)	78.9	77.6	78.1	78.2
Sonata (lin.)	72.2	68.4	70.1	68.9
+LoRA	75.6	71.3	73.5	72.0
+PointTPA	78.4	73.9	76.8	74.6
+PointPiT	78.9	78.0	78.6	78.2

Table 5: Overall analysis of key hyperparameters in PointPiT.

r	mIoU	q	mIoU	λ	mIoU
16	76.5	8	77.6	0.1	78.2
32	78.0	16	78.4	0.2	78.6
64	78.9	32	78.9	0.5	78.9
128	78.7	64	78.6	1.0	78.3
256	78.5	128	78.2	2.0	77.6

Evaluation Metrics. We report mean Intersection over Union (mIoU, %), mean Accuracy (mAcc, %), and overall Accuracy (allAcc, %) for semantic segmentation following prior works [15]. We additionally report the number of trainable parameters to evaluate the parameter efficiency of different adaptation methods.

3.2. Main Results

Quantitative Results. As shown in Table 1, compared to full fine-tuning, PointPiT achieves competitive performance and even exceeds it on Scannet [19] with less than 1% of the backbone’s parameters. Across semantic categories, PointPiT attains the best or second-best results in most cases. Furthermore, Table 2 demonstrates that PointPiT consistently outperforms all representative PEFT methods across diverse scene-level benchmarks. These benchmarks span distinct domains with substantial differences in geometric structure and semantic granularity. Under this domain gap, Under this domain gap, the joint design of SSA and GSO in PointPiT proves especially beneficial.

Qualitative Results. As illustrated in Fig. 3, PointPiT produces more accurate and consistent predictions in complex scene regions compared with FFT, with fewer semantic misclassifications and more coherent object structures. These qualitative results further demonstrate that PointPiT can preserve fine-grained semantic information while maintaining scene-level structural consistency under parameter-efficient adaptation.

3.3. Ablation Study

Effect of SSA and GSO. Table 3 demonstrates the complementary contributions of SSA and GSO. Although each component independently improves performance, their combination achieves further

gains. This indicates that effective scene-level PEFT benefits from both representation adaptation and parameter update perspectives.

Sensitivity to Serialization Configuration. Different serialization configurations result in distinct partition structures, which may introduce unstable adaptation patterns for PEFT methods. By capturing more stable structural patterns and avoiding over-reliance on a specific partition layout, PointPiT maintains consistent performance across different configurations, as shown in Table 4. This suggests that the proposed adaptation strategy improves the generalization ability of scene-level models under varying serialization schemes.

Analysis of Key Hyperparameters. Table 5 shows that $r = 64$, $q = 32$, and $\lambda = 0.5$ achieve the best performance. Smaller r or q limits adaptation capacity or informative updates, while excessive values introduce diminishing returns. Similarly, overly strong partition regularization may suppress useful gradients. These results indicate that PointPiT achieves effective adaptation with moderate model capacity and selective gradient updates, without relying on excessive parameter expansion or partition regularization.

4. CONCLUSION

In this paper, we propose PointPiT, a partition-invariant tuning framework for large-scale scene-level point clouds. We identify the mismatch between serialization-induced partition variations and parameter-efficient adaptation, and address it through scene-aware structural adaptation and gradient subspace optimization. Extensive experiments demonstrate that PointPiT achieves consistent state-of-the-art performance among PEFT methods and competitive or even superior performance to full fine-tuning with less than 1% of the backbone’s parameters. As an early exploration of scene-level PEFT, we expect PointPiT to serve as a strong baseline and provide insights for future research on efficient 3D scene adaptation.

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